

README:

Introduction:

Welcome to the documentation for the Chatbot project. This document provides an overview of the project structure, file descriptions, and explanations for key components.

Overview:

Kiki is made for students like us studying economics. It provides a conversational interface to users, capable of understanding and responding to various user inputs. It can handle general questions like greetings and goodbyes. Specifically, it's good at:

- Solving basic and financial math problems (Duration, NPV, and Future Value).
- Getting news from Google.
- Checking stock summaries on Yahoo.
- Helping with task planning by adding, viewing, and deleting items.
- Searching on Wikipedia.
- Giving date and time updates.
- Providing current weather info from Open-Meteo API.

Setup:

Before running the code, you must install some package. Run the code below in your terminal path:

- ❖ *pip install nltk* (for natural language processing)
- ❖ *pip install numpy* (for complex calculations)
- ❖ *pip install requests* (for making HTTP requests)
- ❖ *pip install sqlite3* (for interacting with the SQLite database)
- ❖ *pip install bs4* (for web scraping)

"run.py" is the file that kicks off the program. You can start it by clicking the button in your code editor or by typing "python run.py" in the terminal.

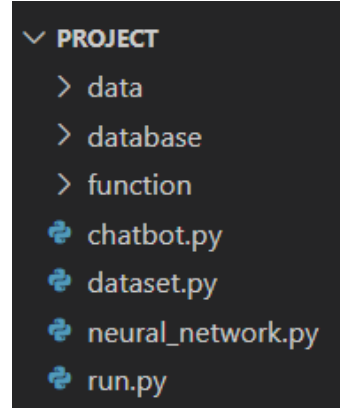
Make sure your terminal path is set to the Kiki chatbot folder (like this: \users\your path\Kiki chatbot>).

Before program starts, you'll be asked to choose the language (for now English and Italian are available).

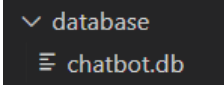
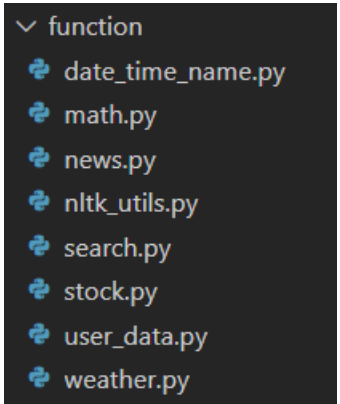
The first time you run the program, you'll see a progress bar displaying the loss and progress of neural networks training. This happens as the program trains the neural networks. Once the training is complete, Kiki will prompt you for your information. After this step, Kiki is ready to answer your questions.

CODE STRUCTURE:

- The `"run.py"` file serves as the main entry point for the project. It initializes the neural network, loads the dataset, trains the model, and runs the chatbot.
- The `"neural_network.py"` file contains the implementation of the neural network class. It handles the initialization, training, and prediction tasks. The class uses a specific architecture with layers and activation functions.
- The `"dataset.py"` file is responsible for loading and preprocessing the dataset used for training the neural network. It parses data, creates bags of words, and provides methods to retrieve relevant information.
- The `"chatbot.py"` file contains the main chatbot class. It integrates functionalities from various modules in the "function" folder

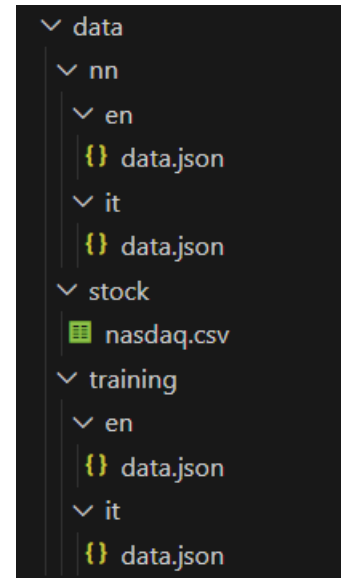


- The folder "function" contains local modules used by the chatbot.
 - The `"date_time_name.py"` file provides functionality for handling date and time information, including the conversion of date values to readable formats.
 - The `"math.py"` contains the "Calculator" class used for financial and simple calculations.
 - The `"news.py"` contains functionality for fetching news articles or headlines from Google News. It uses web scraping techniques (bs4).
 - The `"nltk_utils.py"` file contains the "Word" class, which uses the Natural Language Toolkit (NLTK) library for natural language processing tasks. It provides methods for tokenizing sentences, stemming words using the Porter stemming algorithm, and generating a bag-of-words representation for given patterns and a predefined set of words.
 - The `"search.py"` file contains the "SearchEngine" class, which facilitates searches using the Wikipedia API.
 - The `"stock.py"` file provides the "Stock" class, which provides methods to retrieve stock information using data from the NASDAQ stock screener (in /data/stock/nasdaq.csv).
 - The `"user_data.py"` file contains the "UserData" class, which manages user-related data using an SQLite database ("chatbot.db") in the folder "database".
 - The `"weather.py"` file provides the "WeatherForecast" class, which uses the Open Meteo API to provide current weather information for a given city (default = "Milan"). It includes methods to obtain the geographical position of a city and its weather forecast.



- The folder “data” contains subfolders:
 - “nn” that contains data related to the neural network. In this there are the weights and biases of the trained model stored in a JSON file
 - “stock” which holds a CSV file (“nasdaq.csv”) with names and symbols of companies listed on NASDAQ.
 - “training” that contains data used for training the neural network. This includes datasets used during the model training process. The file JSON (“data.json”) are stored the responses based on predefined intents.

en and it in nn and training folders identify the language of data.



USER GUIDE

To use the Chatbot, follow these steps:

1. Run the Chatbot

Execute “run.py” to start the chatbot or type “python run.py” in the terminal.

2. Language choice.

```
Which language do you speak? (press 'E' for English, 'I' for Italian): e
```

3. Chatbot Interaction

- Provide your name, your age and your hobby to the Chatbot when prompted.

```
Kiki: Hi! What's your name?: Pippo
Kiki: Nice to meet you! How old are you?: 91
Kiki: Which is your hobby?: Eat bones
Kiki: Hello Pippo! How can I help you?
Pippo: |
```

The Chatbot greets the user, asks for their name, and stores user data.

```
Kiki: Hi! What's your name?: Pippo
Kiki: Nice to see you again Pippo! How can I help you?
Pippo:
```

If the user is recognized, a personalized greeting is provided.

- Engage in a conversation with the Chatbot by entering messages.

Examples:

WHEATER

If you ask to Kiki information about the weather, it will retrieve the weather information for the specified city and respond with details about the current temperature range, precipitation probability, and average wind speed.

```
Kiki: Hi! What's your name?: Pippo
Kiki: Nice to see you again Pippo! How can I help you?
Pippo: Hi, Kiki! What's the weather today?
Kiki: Which city are you looking for?: Rome
Kiki: Today, in Rome the temperature is from 1.1 to 12.2.
Precipitation probability is about 0.0% and wind speed is around 6.86 km/h.
```

PLANNING

If you inquire with Kiki about your plans, it will provide details on the number and specific content of plans you've previously added.

You can manage your plans by typing "a" to add a new plan or "d" to delete a plan, followed by the corresponding number from the displayed list. Alternatively, you can choose to leave everything unchanged.

```
Pippo: I have a plan to add
Kiki: You have 0 item(s) of plan:
Kiki: press 'D' to delete a plan, 'A' to add a new plan, 'E' to exit: a
Kiki: What is the plan?: Advanced python exam
Kiki: When is the plan? (dd/mm/yyyy): 13/12/2023
Kiki: Plan added!
Pippo: Do you have any plans?
Kiki: You have 1 item(s) of plan:
Kiki: 1) On 2023-12-13: Advanced python exam
Kiki: press 'D' to delete a plan, 'A' to add a new plan, 'E' to exit: d
Kiki: Which item do you want to delete? (press the number): 1
Kiki: 1) On 2023-12-13 Advanced python exam has been deleted
Pippo: Do you have any plans?
Kiki: You have 0 item(s) of plan:
Kiki: press 'D' to delete a plan, 'A' to add a new plan, 'E' to exit: e
```

DATE AND TIME

Whether you need to know today's date and the current time, Kiki can provide you with the information.

```
Pippo: What day is it?
Kiki: Today is Thursday, December 8, 2023, at 14:50
```

NEWS

Kiki can fetch and present the latest news based on your queries, helping you stay informed and updated.

```
Pippo: What is the latest news?
Kiki: How many items of news are you interested to see?: 5
Kiki: Top 5 news:

UNLV shooting: Vegas shooter did not appear to target students
1 hour ago
-----
UNLV shooting suspect was professor who had applied for job: AP source
11 hours ago
-----
UNLV students got alerts to run, hide, fight during active shooting
6 hours ago
-----
Mental health professional talks coping, healing after UNLV tragedy
4 hours ago
-----
First on CNN: Federal prosecutors file new criminal case against Hunter Biden
1 hour ago
-----
```

STOCK

If you ask to Kiki info about stocks, it can provide information about companies listed on NASDAQ. You can have details about a specific stock by mentioning its name, and Kiki will retrieve relevant stock information such as the current stock values and related statistics.

```
Pippo: Can you tell me how a stock is priced?
Kiki: Sure! Could you please let me know the name of the company you are looking for?:Apple
Kiki: For Apple (AAPL), here is the summary:

Open:                191.41                |Bid:                190.87 x 1200                |
-----
Ask:                190.88 x 800                |Day's Range:        190.27 - 191.50                |
-----
52 Week Range:        124.17 - 198.23                |Volume:            9,812,811                |
-----
Avg. Volume:          58,702,353                |Market Cap:         2.96T                |
-----
Beta (5Y Monthly):    1.30                |PE Ratio (TTM):     31.05                |
-----
EPS (TTM):            6.13                |Earnings Date:      Jan 31, 2024 - Feb 05, 2024    |
-----
Forward Dividend & Yield:0.96 (0.50%)            |Ex-Dividend Date:   Nov 10, 2023                |
-----
```

SEARCH

If you ask Kiki to search for information, it will use Wikipedia to find relevant details based on your query.

```
Pippo: Can you search for information for me?
Kiki: Could you provide specific details about what you're seeking?: apple
Kiki:
Here is what I found:
An apple is a round, edible fruit produced by an apple tree (Malus spp., among them the domestic or orchard apple; Malus domestica). Apple trees are cultivated worldwide and are the most widely grown species in the genus Malus. The tree originated in Central Asia, where its wild ancestor, Malus sieversii, is still found. Apples have been grown for thousands of years in Asia and Europe and were introduced to North America by European colonists. Apples have religious and mythological significance in many cultures, including Norse, Greek, and European Christian tradition.
Apples grown from seed tend to be very different from those of their parents, and the resultant fruit frequently lacks desired characteristics. For commercial purposes, including botanical evaluation, apple cultivars are pro...

1) An apple is an edible fruit.
Apple, Apples or APPLE may also refer to:

Apple Inc., an American mul...
2) Apple Inc. is an American multinational technology company headquartered in Cupertino, California. A...
3) Apple Music is a music, audio and video streaming service developed by Apple Inc. Users select music...
4) Apple Records is a record label founded by the Beatles in 1968 as a division of Apple Corps Ltd. It ...
```

MATH

You can interact with Kiki by entering mathematical expressions or choosing financial formulas. Kiki will provide you with the results or guide you through financial calculations based on your input.

```
Pippo: Can you do calculations for me?
Kiki: Enter your expression (press 'F' for financial math): 10+3*3
Kiki: The result is 19
```

For financial math, it can calculate:

- Duration of an investment
- Net Present Value (NPV) of an investment
- Future Value of an investment

For example, the calculation of the NPV:

```
Pippo: Can you do calculations for me?
Kiki: Enter your expression (press 'F' for financial math): f
Kiki: What do you want to calculate? (press 'D' for duration, 'N' for NPV, 'F' for future value): n
Kiki: Enter the initial investment: 99
Kiki: Enter the number of years: 2
Kiki: Enter the cost of capital (in %): 4
Kiki: Enter the cash flow in year 1: 5
Kiki: Enter the cash flow in year 2: 105
Kiki: The NPV of the investment is: 2.89
```

GOODBYE

If you say “bye” or also simply “exit” to Kiki, it will respond with a farewell message and exit the conversation.

```
Pippo: Bye  
Kiki: Do you want to exit? (Y/N): y  
Kiki: Goodbye!
```

Beta functions:

While these functions have been developed, it's important to note that they may contain a few bugs as they have not yet undergone testing.

LANGUAGE

Manages language-specific texts for a chatbot.

```
manager = LanguageManager('en')  
greeting_message = manager.get_text('GREETING_MESSAGE', name='John')  
print(greeting_message) # Output: "Hello John! How can I help you?"
```

IMPROVE

The code aims to enhance the dataset used by Kiki by refining the tags or categories associated with unanswered questions.

```
Diego: Nihao  
Kiki: Sorry, can't understand you  
Diego: bye  
Kiki: Do you want to exit? (Y/N): y  
Kiki: Goodbye!  
would you like to help us to improve the data? (Y/N): y  
This is the beta version of the data improving function...  
Is 'Nihao' a type of greeting? (Y/N): y  
Thank you!
```

```
Diego: nihao!  
Kiki: Hi there! Thanks for asking. I'm good, and you?
```


Theoretical Insights into Neural Networks

Some information about **backpropagation** algorithm:

Chain Rule:

In [calculus](#), the **chain rule** is a [formula](#) that expresses the [derivative](#) of the [composition](#) of two [differentiable functions](#) f and g in terms of the derivatives of f and g . More precisely, if $h = f \circ g$ is the function such that $h(x) = f(g(x))$ for every x , then the chain rule is, in [Lagrange's notation](#),

$$h'(x) = f'(g(x))g'(x).$$

or, equivalently,

$$h' = (f \circ g)' = (f' \circ g) \cdot g'.$$

The chain rule may also be expressed in [Leibniz's notation](#). If a variable z depends on the variable y , which itself depends on the variable x (that is, y and z are [dependent variables](#)), then z depends on x as well, via the intermediate variable y . In this case, the chain rule is expressed as

$$\frac{dz}{dx} = \frac{dz}{dy} \cdot \frac{dy}{dx},$$

and

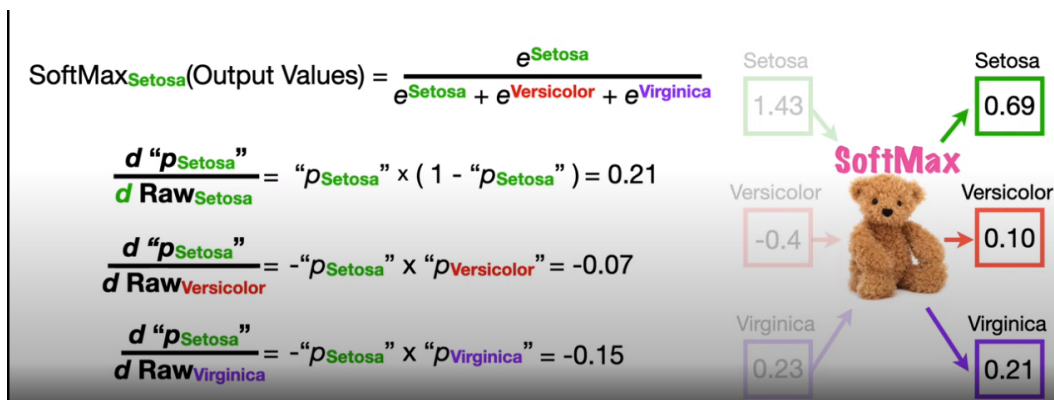
$$\left. \frac{dz}{dx} \right|_x = \left. \frac{dz}{dy} \right|_{y(x)} \cdot \left. \frac{dy}{dx} \right|_x,$$

for indicating at which points the derivatives have to be evaluated.

In [integration](#), the counterpart to the chain rule is the [substitution rule](#).

$$\frac{d \text{CE}_{\text{Setosa}}}{d b_3} = \frac{d \text{CE}_{\text{Setosa}}}{d "p_{\text{Setosa}}"} \times \frac{d "p_{\text{Setosa}}"}{d \text{Raw}_{\text{Setosa}}} \times \frac{d \text{Raw}_{\text{Setosa}}}{d b_3}$$

Derivative of SoftMax:



derivative of loss (cross entropy) with respect to activation function (SoftMax)

$$\frac{d \text{CE}_{\text{Setosa}}}{d b_3} = \frac{d \text{CE}_{\text{Setosa}}}{d "p_{\text{Setosa}}"} \times \frac{d "p_{\text{Setosa}}"}{d \text{Raw}_{\text{Setosa}}} \times \frac{d \text{Raw}_{\text{Setosa}}}{d b_3}$$

$$= \frac{-1}{"p_{\text{Setosa}}"} \quad \leftarrow \text{And thus, the derivative of the Cross Entropy with respect to the Predicted "probability" for Setosa is...}$$

The neural network's output is a list with a length matching that of the tags in data.json. Each element in the list represents the percentage probability associated with its corresponding tag.

```
message Hi
tags ['goodbye', 'greeting', 'ok']
predicted output [[0.03461942 0.92016188 0.04521869]]
```

Video reference:

https://www.youtube.com/watch?v=xBEh66V9gZo&list=PLblh5JKOoLUixGDQs4LFFD--41Vzf-ME1&index=13&ab_channel=StatQuestwithJoshStarmer

https://www.youtube.com/watch?v=M59JEIEPgIg&list=PLblh5JKOoLUixGDQs4LFFD--41Vzf-ME1&index=11&t=362s&ab_channel=StatQuestwithJoshStarmer

Next Step:

Indeed, while extending Kiki's knowledge("data.json") allows to enrich the comprehension, there's a trade-off. Much information is contained in data.json, longer will be the training time, and in extreme cases, it can become quite lengthy.

To address this and handle more intricate challenges, a vanilla Artificial Neural Network (ANN) may not be the optimal solution. For enhanced efficiency and tackling complex problems, we are exploring advanced techniques like Recurrent Neural Networks (RNN) and even more sophisticated ones like Long Short-Term Memory (LSTM) networks. These advancements aim to make chatbot smarter and more robust in natural language processing.

Expect to find these features in Kiki 2.0

